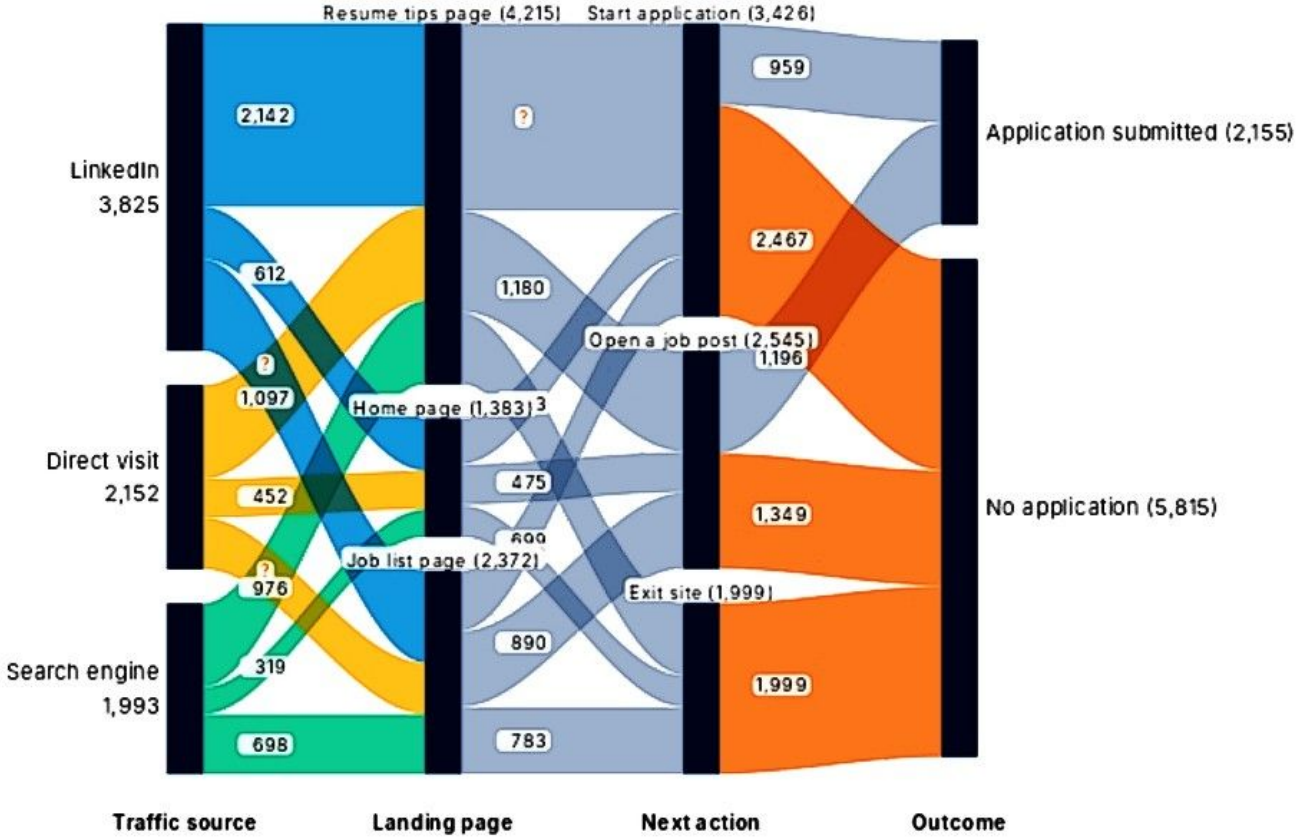
DSA06 - Data Handling and Visualization | Activity 6: Website Flow Sankey Design | Slot C

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Scenario: The analytics team of a job portal tracked 7,970 visits last month through four stages: traffic source, landing page, next action and outcome. The Sankey diagram below shows the number of visits on every flow. Three flows are hidden and marked "?".



(a) Using conservation of flow (what enters a node equals what leaves it) find the three hidden flows, showing working: LinkedIn to Job list page; Direct visit to Job list page; Resume tips page to Start application. State the rule you used. [4]

(b) Compute: (i) the overall conversion rate, i.e. Application submitted as a percentage of all visits; (ii) the largest single flow into "No application" or "Exit site"; (iii) the percentage of "Job list page" visits that go on to "Start application". [4]

(c) Design the final Sankey: give four design decisions (node order to reduce crossings, colour rule, labels, what to highlight) and justify each in one line using this data. [4]

(d) The Sankey is to be printed with 1 cm of node height = 400 visits. Compute the height in cm of each landing-page node (Home page, Job list page, Resume tips page). [2]

(e) Write R code: a links data frame for the first four flows leaving "Direct visit" and "Search engine" in the table order of the diagram, a nodes data frame, and the networkD3::sankeyNetwork() call (or plotly) that draws the whole diagram. [4]

(f) State two situations in which a Sankey diagram misleads or becomes unreadable, and name one alternative chart for each. [2]

Activity-6

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Course :- Data Handling & Visualization

Code :- DSA0601

Part-A Find the hidden flows.

Total visits = 7,970

1) Hidden flow = Total entering a node - known flows.

3825 - 2142 - 612 - 1097

= -26

2) Total visits = 2,152

2152 - 1097 - 452 = 603

3) Hidden flow = Total entering Page - other outgoing flows.

* The Page has 4215 visits.

Part-B:- Calculations

(i) Overall conversion rate.

$$\frac{2155}{7970} \times 100 = \boxed{27.04\%}$$

ii) largest single flow into two applications.

* 2,467 visits, from start application

iii) Joblist Page → StartApplication

2372

Part-C - final sankey Diagram:

Design:

* Traffic source → Landing Page → next action → outcome.

Part-D - Node Heights

Lem = 400 visits

formula:

Height in Cm = $\frac{\text{No of visits}}{400}$

Part E:

R code.

```
library(networkD3)
Links <- Dataframe(
        source = c(0,0,0,1,
                   c(3,4,5,3),
                   c(2142, 612, 1071, 329))
nodes <- Data.frame(
        name = c())
Links = links,
nodes = nodes,
Source = "source",
Target = "Target",
value = "value",
fontsize = 12,
nodewidth = 30)
```

Part-F: Limitations of Sankey Diagrams.

* The difficult of flows. An alternative is a table or grouped bar chart.